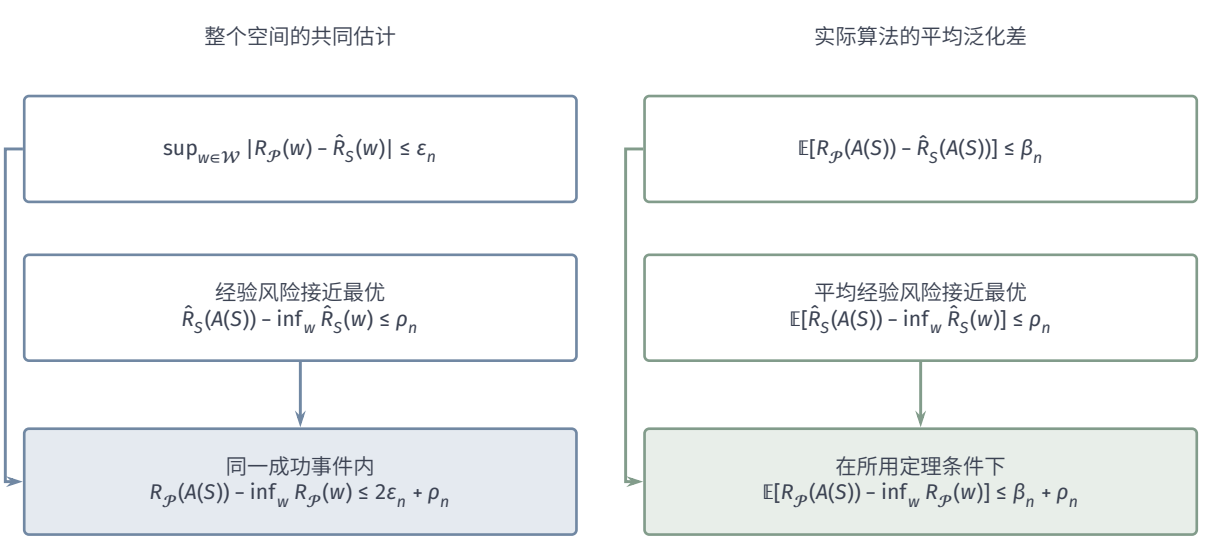
整个空间的共同估计
$\sup_{w \in \mathcal{W}} |R_{\mathcal{P}}(w) - \hat{R}_S(w)| \le \varepsilon_n$
经验风险接近最优
$\hat{R}_S(A(S)) - \inf_w \hat{R}_S(w) \le \rho_n$
同一成功事件内
$R_{\mathcal{P}}(A(S)) - \inf_w R_{\mathcal{P}}(w) \le 2\varepsilon_n + \rho_n$
实际算法的平均泛化差
$\mathbb{E}[R_{\mathcal{P}}(A(S)) - \hat{R}_S(A(S))] \le \beta_n$
平均经验风险接近最优
$\mathbb{E}[\hat{R}_S(A(S)) - \inf_w \hat{R}_S(w)] \le \rho_n$
在所用定理条件下
$\mathbb{E}[R_{\mathcal{P}}(A(S)) - \inf_w R_{\mathcal{P}}(w)] \le \beta_n + \rho_n$